



PROJECT-BASED LEARNING

PHASE-I EVALUATION

InternMatch

DSA & OOP Based Internship Recommendation and Skill-Gap Analysis System

Team ID: DSCPP-III-2026-T024

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T024
Semester: 3rd
Sections: ML1, ML2, Cyber Security
Domain: Internship Recommendation

Team Members

1. Jigyasa Rana – 2510380058
ML2 – Team Lead
2. Masoom Singh – 2510370548
ML1
3. Kashvi Pandey – 2510012207
Cyber Security

Mentor

Dr. Siddhant Thapliyal

Interactions completed: 5

Problem Statement & Motivation

Problem Statement

- ▶ Going through a large database manually takes a lot of time.
- ▶ Checking eligibility criteria and required skills by yourself can lead to mistakes.
- ▶ Students face difficulty in finding the right internship according to their skills.

Motivation

- ▶ Easier internship hunt.
- ▶ Automatic eligibility checking and skill match percentage.
- ▶ Shows the skill gaps.
- ▶ Recommends the best opportunity based on skill match percentage.

Target Users

- ▶ Students or people looking for internships.
- ▶ Placement mentors.

Project Objectives

Objectives

1. Fetch real-time internship data from Arbitnow.com.
2. Take student details such as CGPA, branch and skills as input.
3. Check whether the student is eligible based on CGPA, branch and skills.
4. Calculate match percentage by comparing student skills with required internship skills and rank high-match internships.
5. Identify the missing skills.

Proposed Solution

Proposed Approach

1. Student enters their details at runtime.
2. API fetches real-time internship data.
3. Extract data from JSON.
4. Check the eligibility criteria.
5. Compare required internship skills with the student's skills.
6. Calculate match percentage and find the skill gap.
7. Recommend the best internship in ranked order.

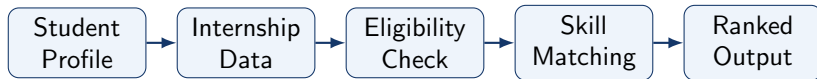
Key Features

- ▶ Student profile input
- ▶ Real-time internship data fetching
- ▶ Automatic eligibility checking
- ▶ Match percentage
- ▶ Skill gap
- ▶ Ranking the best opportunities

Integration of PBL Course Concepts

Course			Concepts Applied	Project Module
Data Structures in C++			Linked List, Linear Searching and Sorting	Internship Data & Matching
OOPs with C++			Classes and Objects, Constructors, Encapsulation and Inheritance	Student & Internship
API Integration with JSON			Fetching real-time internship data and extracting data from JSON.	Internship Data
Algorithms			Eligibility checking, skill comparison, match calculation and ranking	Matching Module

System Architecture / Workflow



Major Components

- ▶ Student Class – stores student details
- ▶ Internship Class – stores internship details fetched using API
- ▶ Matching Engine – compares student and internship details
- ▶ Linked List – stores internship

objects

Step-by-Step Data Flow

- ▶ Internship data is fetched in real time.
- ▶ Data is converted into internship objects.
- ▶ Objects are stored in a Linked List.
- ▶ Student enters branch, CGPA and skills.

▶ Eligibility and skills are checked.

Technology Stack & Methodology

Technology Stack

- ▶ Programming: C++
- ▶ Data Structures: C++
- ▶ OOP: C++
- ▶ API: Arbitnow
- ▶ HTTP Requests: libcurl
- ▶ JSON: nlohmann/json
- ▶ IDE: Visual Studio Code
- ▶ Version Control: Git / GitHub

Development Methodology

1. Problem identification
2. Defining system requirements
3. Finalizing workflow
4. Choosing data structures
5. Making project modules
6. Testing the system

Initial Progress & Team Contribution

Progress Achieved

- ▶ Problem identified
- ▶ System requirements discussed
- ▶ Workflow finalized
- ▶ Student profile details decided
- ▶ API and JSON approach planned
- ▶ Matching logic planned
- ▶ Initial implementation started

Individual Contribution

Member	Role	Contribution
Jigyasa	Team Lead	Planning, requirements and project coordination
Masoom	Developer	DSA logic, eligibility and skill matching
Kashvi	Developer	Data preparation, testing and documentation

Roadmap & Expected Outcomes

Roadmap

Phase-I (Current)

- ▶ Fetch real-time data
- ▶ Student & Internship classes

Phase-II

- ▶ Implement Linked List
- ▶ Check eligibility based on CGPA and branch
- ▶ Rank internships

Phase-III

- ▶ Full integration
- ▶ Create menu-driven interface
- ▶ Test with multiple student profiles
- ▶ Validate input
- ▶ Final documentation

Expected Outcomes

- ▶ Internship suggestions
- ▶ Eligibility check
- ▶ Match percentage
- ▶ Skill gaps
- ▶ Ranked results

References

1. libcurl
2. nlohmann/json
3. Arbitnow API

Thank You

Questions & Suggestions